



In 1D, the WKB approximation for the wavefunction (at order $\hbar^1$) is given by

$$\psi(x) = \frac{1}{\sqrt{p(x)}} \left[A e^{\frac{i}{\hbar} \int^x p(x') dx'} + B e^{-\frac{i}{\hbar} \int^x p(x') dx'} \right], \quad (1)$$

where $p(x) = \sqrt{2m(E - V(x))}$ is the classical momentum. Note that A and B are constants determined by boundary conditions. They describe left and right moving waves, respectively.

For a bound state, the energy of the particle E is less than the potential at some point, which means that there are classically forbidden regions where $p(x)$ becomes imaginary. The point at which this happens is called a turning point, and the WKB approximation no longer holds right at that intersection. To resolve this, one matches the allowed and forbidden region solutions, which in the case of a bound state, leads to a quantization condition for the wavefunction (that is, we find standing wave solutions when the energy is below the potential barriers). Note that classically, the particle would be trapped between the turning points. In other words, classically, the nodes of the wavefunction must coincide with the turning points. In quantum mechanics, however, the particle can tunnel through the barriers even if the probability density still decays exponentially fast outside of it (the “evanescent wave”), but the wavefunction of the bound state must still describe a standing wave.

The matching procedure and therefore the quantization conditions depend on what potential we have. The case of the infinite square well is, however, the easiest one, so we derive it below. The matching procedure will be different if our sharp walls are finite and different still if the walls are “smooth”. Sakurai’s discussion is concerned with the latter case, which is solved by assuming the potential can be linearized near the turning points. While a bit messy, the end result is very similar to the previous cases and they all become more and more alike for larger quantum numbers n (of course, they must all asymptote to the classical limit).



HW 3.1: In this question we will derive the quantization condition for the infinite square well and derive its energy levels.

a) Consider the infinite square well potential

$$V(x) = \begin{cases} V_0 & \text{if } 0 < x < a, \\ \infty & \text{otherwise.} \end{cases} \quad (2)$$

What are the turning points of this potential? Using the wavefunction form in Eq. (1) and the boundary conditions for a infinite square well, show that the quantization condition for this potential is just

$$\int_{x_0}^{x_1} dx p(x) = n\pi\hbar, \quad n = 1, 2, 3, \dots \quad (3)$$

where the integration is taken over a half cycle of the particle, which is defined as the path between one turning point, x_0 , to the other, x_1 . You can choose either complex exponentials or trig functions to solve this problem.

b) Use the quantization condition to derive the energy levels of the infinite square well with $V_0 = 0$, and compare with the exact solution (namely, $E_n = n^2\pi^2\hbar^2/2ma^2$).

c) Recall we used defined the quantum potential in our notes: $Q(x) = -\frac{\hbar^2}{2m} \sqrt{p(x)} \frac{d^2}{dx^2} \left(\frac{1}{\sqrt{p(x)}} \right)$. Calculate $Q(x)$ inside the infinite square well potential.

d) Imagine now that instead of a flat well ($V(x) = V_0$), we have an arbitrary but continuous potential that satisfies $V(x) < E$ inside the infinite well. Do we expect the WKB approximation

to be exact in this case? If not, under what conditions do we expect the WKB approximation to work well?



HW 3.2: Now we look into a couple of other potentials. Given a quantization condition, we will find the turning points and then derive the energy levels of the system.

a) Consider the smooth wall version of the infinite square well, given by the potential

$$V(x) = \begin{cases} V_0 & \text{if } -a < x < a, \\ V_0 \left(\frac{x}{a}\right)^\gamma & \text{if } x \geq a, \\ V_0 \left(\frac{-x}{a}\right)^\gamma & \text{if } x \leq -a, \end{cases} \quad (4)$$

where $V_0 < E$ is a positive constant and γ is a positive integer. This potential has the same shape as the infinite square well, but with smoother walls. Because of it, the nodes of the wavefunction are inside the potential and the quantization condition becomes (same as Eq. 2.254 in Sakurai)

$$\int_{x_0}^{x_1} p(x) dx = \left(n + \frac{1}{2}\right) \pi \hbar, \quad n = 0, 1, 2, \dots \quad (5)$$

Find the turning points x_0 and x_1 of this potential. Use this quantization condition to derive the energy levels of this potential, and compare with the infinite square well case.

b) Finally, consider an atom with a single electron, with

$$V(r) = -\frac{\alpha Z}{r}, \quad (6)$$

where α is the fine structure constant and Z is the atomic number. We can imagine $r = 0$ as a hard wall (first turning point) and the second turning point to be some value a such that $E = V(a)$, that is $a = -\frac{\alpha Z}{E}$. The quantization condition for this potential is given by

$$\int_{x_0}^{x_1} p(r) dr = \left(n - \frac{1}{4}\right) \pi \hbar, \quad n = 1, 2, \dots \quad (7)$$

Find the turning points x_0 and x_1 of this potential. Use this quantization condition to derive the energy levels of the atom. Do we expect the solution for $n > 1$ to be a good approximation for the Hydrogen atom energy levels? How about the $n = 1$? Compare the latter with the exact solution for ground state of the hydrogen atom.

Solutions to HW 3.1: a) The turning points of the infinite square well are at $x_0 = 0$ and $x_1 = a$, where the potential goes to infinity. Let us rewrite the wavefunction in terms of some generic phase, which is determined by our WKB solution:

$$\psi(x) = \frac{1}{\sqrt{p(x)}} \left[A e^{\frac{i}{\hbar} \phi(x)} + B e^{-\frac{i}{\hbar} \phi(x)} \right], \quad \phi(x) = \int^x p(x') dx' = \sqrt{2m(E - V_0)} x. \quad (8)$$

First we can apply the boundary conditions $\psi(0) = 0$ to find that $A = -B$. This leads to a sine function form for our solution:

$$\psi(x) = \frac{2iA}{\sqrt{p(x)}} \sin\left(\frac{\phi}{\hbar}\right) = \frac{2iA}{\sqrt{p(x)}} \sin\left(\frac{\sqrt{2m(E - V_0)} x}{\hbar}\right). \quad (9)$$

Then, applying the boundary condition $\psi(a) = 0$ leads us to the quantization condition $\sin(\phi(a)/\hbar) = 0$, which is satisfied when $\phi(a) = n\pi\hbar$:

$$\phi(a) = \sqrt{2m(E - V_0)} a = n\pi\hbar. \quad (10)$$

Note that we are assuming that $E > V_0$ so that the inner region of the well is classically allowed.



This was the easiest quantization condition to derive. The calculation is analogous to what you did the first time you encountered this potential. The sharp boundary and infinitely tall wall makes the “classically forbidden” region completely inaccessible to the wavefunction, so the nodes of the wavefunction must coincide with the turning points.

b) Rearranging the equation above, we find

$$E_n = V_0 + \frac{n^2 \pi^2 \hbar^2}{2ma^2}, \quad (11)$$

which up to the overall additive constant of the potential V_0 is exactly the same as the exact solution for the infinite square well.

c) Inside the infinite square well, we have $p(x) = \sqrt{2m(E - V_0)}$, which is assumed to be constant. Therefore, the quantum potential vanishes and we should not be surprised that the WKB approximation is exact in this case.

d) If the potential varies, then the WKB approximation inside the well need not be exact. The quantization condition is still trivial (in the sense that we can derive it as we did above with the vanishing wavefunction at the boundaries), but the energy levels may be different from the infinite square well case. The validity condition for the WKB is that the potential should not vary too rapidly. As shown in the notes, this is equivalent to having $|d\lambda/dx| \ll 1$, where $\lambda(x) = h/p(x)$ is the local wavelength of the particle.

Solutions to HW 3.2: a) The turning points are defined by the region where $E = V(x)$, which leads to

$$E = V_0 \left(\frac{x}{a}\right)^\gamma \implies x_1 = -x_0 = a \left(\frac{E}{V_0}\right)^{1/\gamma}. \quad (12)$$

That is, the turning points are now in a region where the potential is varying. Two things to note 1) the larger the energy of our particle, the larger is the interval between the turning points, and 2) for $\gamma \rightarrow \infty$, we recover the turning points of the infinite square well case, as expected.

Let us define for convenience dimensionless measures of distance and energy:

$$y = \frac{x}{a}, \quad \epsilon = \frac{E}{V_0}. \quad (13)$$

In this case, our dimensionless potential is just

$$v(y) = V(x)/V_0 = \begin{cases} |y|^\gamma & |y| \geq 1, \\ 1 & |y| < 1, \end{cases} \quad (14)$$

and the left-hand side of the quantization condition becomes

$$a\sqrt{V_0} \int_{-\epsilon^{1/\gamma}}^{\epsilon^{1/\gamma}} \sqrt{2m(\epsilon - v(y))} dy = a\sqrt{2mE} \int_{-\epsilon^{1/\gamma}}^{\epsilon^{1/\gamma}} \sqrt{1 - \frac{v(y)}{\epsilon}} dy = 2a\sqrt{2mE} \int_0^{\epsilon^{1/\gamma}} \sqrt{1 - \frac{v(y)}{\epsilon}} dy \quad (15)$$

where we used the fact that the potential is symmetric around 0. We can then write the quantization condition as

$$I(\epsilon, \gamma) \stackrel{!}{=} \frac{(n + \frac{1}{2}) \pi \hbar}{2a\sqrt{2mE}}. \quad (16)$$

where we defined the integral

$$I(\epsilon, \gamma) \equiv \int_0^{\epsilon^{1/\gamma}} \sqrt{1 - \frac{v(y)}{\epsilon}} dy = \int_0^1 \sqrt{1 - \frac{1}{\epsilon}} dy + \int_1^{\epsilon^{1/\gamma}} \sqrt{1 - \frac{y^\gamma}{\epsilon}} dy \quad (17)$$

$$= \sqrt{1 - \frac{1}{\epsilon}} + \epsilon^{1/\gamma} I'(\epsilon, \gamma), \quad (18)$$

with

$$I'(\epsilon, \gamma) = \int_{\epsilon^{-1/\gamma}}^1 \sqrt{1 - u^\gamma} du. \quad (19)$$

Note how we had to account for the two functional forms of the potential inside the classically allowed region. The Eq. (16) is an implicit equation for the energy levels of the system, which can be solved numerically for any given γ .



Note that $I'(\epsilon, \gamma \rightarrow 0) \rightarrow 0$ is a bad limit for $I(\epsilon, \gamma)$ since our turning points go off to infinity. On the other hand, for very steep potential walls, $I'(\epsilon, \gamma \rightarrow \infty) \rightarrow 0$, and

$$I(\epsilon, \gamma \rightarrow \infty) \rightarrow \sqrt{1 - \frac{1}{\epsilon}} = \frac{\sqrt{E - V_0}}{\sqrt{E}} \implies E_n = \frac{(n + \frac{1}{2})^2 \pi^2 \hbar^2}{2m(2a)^2} + V_0, \quad (20)$$

We recovered the turning points for the infinite square well, but we did not quite recover the same energy levels (the form of the solution is the same, but in one case we have a n dependence and in the other we have a $(n + 1/2)$ dependence!) This is expected because the derivation of this quantization condition *assumes* a smooth wall potential (in particular, it assumes that the potential can be linearized near the turning points, Sakurai Sec. 2.5.3 and 2.5.4), which our limit of $\gamma \rightarrow \infty$ breaks down.

Suppose that instead we got rid of the flat bottom of the potential and just had

$$V(x) = V_0 |x/a|^\gamma \quad \text{or equivalently} \quad v(y) = |y|^\gamma, \quad (21)$$

for all x ($y = x/a$). In this case, the turning points are given by $x_0 = -a\epsilon^{1/\gamma}$ and $x_1 = a\epsilon^{1/\gamma}$, and the quantization condition becomes

$$\int_{x_0}^{x_1} p(x) dx = 2a\sqrt{2mE} \int_0^{\epsilon^{1/\gamma}} \sqrt{1 - \frac{y^\gamma}{\epsilon}} dy = 2a\sqrt{2mE} \epsilon^{1/\gamma} I(\gamma) \stackrel{!}{=} \left(n + \frac{1}{2}\right) \pi \hbar. \quad (22)$$

where we defined the integral

$$I(\gamma) = \int_0^1 \sqrt{1-u^\gamma} du, \quad (23)$$

which is just a number that depends on γ (the energy dependence is entirely in the prefactor $\sqrt{E} \epsilon^{1/\gamma} = E^{\frac{1}{2} + \frac{1}{\gamma}} / V_0^{1/\gamma}$). This is a much simpler equation to solve¹, since the integral does not depend on the energy anymore, and we can just solve for the energy levels:

$$E_n = V_0 \left[\frac{(n + \frac{1}{2}) \pi \hbar}{2a I(\gamma) \sqrt{2mV_0}} \right]^{\frac{2\gamma}{\gamma+2}}. \quad (24)$$

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If we insist in calculating the integral, it can be done, but the general case requires special functions. The more straightforward cases are $\gamma = 1$:

$$I(1) = \int_0^1 \sqrt{1-y} dy = \int_1^0 \sqrt{u} du = \frac{2}{3}, \quad (25)$$

and $\gamma = 2$:

$$I(2) = \int_0^1 \sqrt{1-y^2} dy = \int_0^{\pi/2} \cos^2 \theta d\theta = \left(\int_0^{\pi/2} d\theta + \int_0^{\pi/2} \cos 2\theta d\theta \right) = \left(\frac{\pi}{4} + 0 \right) = \frac{\pi}{4}. \quad (26)$$

The general case involves the beta function, which is defined as $B(x, y) = \int_0^1 t^{x-1} (1-t)^{y-1} dt$. We can rewrite our integral as

$$I(\gamma) = \int_0^1 \sqrt{1-y^\gamma} dy = \int_0^1 (1-t)^{1/2} t^{1/\gamma-1} dt = \frac{1}{\gamma} B\left(\frac{1}{\gamma}, \frac{3}{2}\right) = \frac{1}{\gamma} \frac{\Gamma(1/\gamma) \Gamma(3/2)}{\Gamma(1/\gamma + 3/2)}, \quad (27)$$

where we used the change of variables $t = y^\gamma$ and the relation of beta to gamma functions, $B(x, y) = \frac{\Gamma(x) \Gamma(y)}{\Gamma(x+y)}$.

The case of $\gamma = 2$ is particularly interesting, since it corresponds to the harmonic oscillator potential, and we can find the energy levels in closed form:

$$E_n = \left(\frac{(n + 1/2) \pi \hbar}{2a I(2)} \right)^1 \left(\frac{V_0^2}{ma^2} \right)^{1/4} = \left(n + \frac{1}{2} \right) \hbar \sqrt{\frac{2V_0}{ma^2}}. \quad (28)$$

where we can readily read off the frequency of the oscillator $\omega = \sqrt{2V_0/(ma^2)}$

The case of $\gamma = 1$ is the one that Sakurai uses to solve the “bouncing neutron” problem, where the potential is linear in the height of the neutron above the ground, and the energy levels are given by

$$E_n = \left(\frac{(n + 1/2) \pi \hbar}{2a I(1)} \right)^{2/3} \left(\frac{V_0^2}{ma^2} \right)^{1/3} = \left(\frac{9\pi^2 \hbar^2 V_0^2}{32ma^4} \right)^{1/3} (n + 1/2)^{2/3}. \quad (29)$$

b) Now to the Coulomb potential. We are glossing over a lot of the details of the Hydrogen atom here, but the idea is that we are modeling the radial direction as a 1D problem in r , with the imposed requirement that $r > 0$ (it must be the case if we have any hope of interpreting it as a radius). The turning points are defined by the region where $E = V(r)$. For small positive

¹In retrospect, this is the problem we should have solved instead.

$r = 0^+$, the potential goes to negative infinity, but at 0^- it goes to positive infinity, so effectively, we can imagine a hard wall at $r = 0$. For larger r , the potential is becoming less negative, so there is a second turning point at $r = a$ where $E = V(a)$, that is, our non-trivial turning point is given by the radial distance

$$a = -\frac{\alpha Z}{E} = \frac{\alpha Z}{|E|} \quad (30)$$

Note that the energy of the particle must be negative for there to be two turning points, which is consistent with the fact that we are looking for bound states (the potential is negative for $r > 0$ and goes to zero at infinity). We can interpret a as some kind of “classical radius” of the atom.



The definition of the fine structure constant is $\alpha = \frac{e^2}{4\pi\epsilon_0\hbar c}$. In both SI and natural units, it is a dimensionless constant that characterizes the strength of the electromagnetic interaction.

The energy levels are found by integrating the momentum between the turning points. Using $|E| = -E$ and $|V(r)| = -V(r)$, we have

$$\int_{r_0}^{r_1} \sqrt{2m(-|E| + |V(r)|)} dr = \sqrt{2m|E|} \int_0^a \sqrt{\frac{\alpha Z}{|E|r} - 1} dr \quad (31)$$

$$= \sqrt{2m|E|} \int_0^a \sqrt{\frac{a}{r} - 1} dr = \sqrt{2m|E|} a I, \quad (32)$$

where $I = \int_0^1 \sqrt{\frac{1}{y} - 1} dy = \pi/2$ is just a number. Again, we reduced our bound state problem to solving an integral. Using our quantization condition we can find the energy levels:

$$E_n = -\frac{2m\alpha^2 Z^2}{\hbar^2(n - 1/4)^2\pi^2} I^2 = \left(-\frac{m\alpha^2 Z^2}{2n^2\hbar^2}\right) \frac{n^2}{(n - 1/4)^2}. \quad (33)$$

The quantity in the parenthesis is just the exact solution to the hydrogen energy levels. For large $n \rightarrow \infty$, our WKB solution asymptotes to the exact solution. For $n = 1$ we find $E_1 = 1.78E_1^{\text{exact}}$, which is not great, but also not terrible given the simplicity of our approach (our radial integral does not even account for the the usual r^2 factor in the measure of the radial integral! This problem is a 3D one, and our WKB setup was always in 1D).



Note that for $n > 1$ there is additional problem with our approach. For these modes, the centrifugal barrier force can become sizeable:

$$V(r) = -\frac{\alpha Z}{r} + \frac{\hbar^2 l(l+1)}{2mr^2}, \quad (34)$$

where l is the orbital angular momentum quantum number. This means that the effective potential has a different shape. The quantization condition is different still because now the wall at $r = 0$ is not a hard wall, but rather a smooth $1/r^2$ potential. I encourage you to try this version of the problem for yourself using the quantization condition from our previous double-smooth wall problem (now shifted to start at $n = 1$ instead of $n = 0$), and see what you get:

$$\int_{x_0}^{x_1} p(r) dr = \left(n - \frac{1}{2}\right) \pi \hbar, \quad n = 1, 2, \dots \quad (35)$$